

ENV 797 - Time Series Analysis for Energy and Environment Applications | Spring 2026

Assignment 6 - Due date 02/27/26

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Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., “LuanaLima_TSA_A06_Sp26.Rmd”). Then change “Student Name” on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

R packages needed for this assignment: “ggplot2”, “forecast”, “tseries” and “sarima”. Install these packages, if you haven’t done yet. Do not forget to load them before running your script, since they are NOT default packages.

```
knitr::opts_chunk$set(tidy = TRUE, tidy.opts = list(width.cutoff = 20))  
library(cowplot)  
library(forecast)  
library(sarima)  
library(tidyverse)  
library(tseries)
```

This assignment has general questions about ARIMA Models.

Q1

Describe the important characteristics of the sample autocorrelation function (ACF) plot and the partial sample autocorrelation function (PACF) plot for the following models:

- AR(2)

Answer: In this second-order AR model, the current value depends on two previous values. While ACF will gradually decay, PACF will cut off with lag 2.

- MA(1)

Answer: In this first-order MA model, the current deviation from mean depends on one previous deviation from mean. While PACF will gradually decay, ACF will cut off with lag 1.

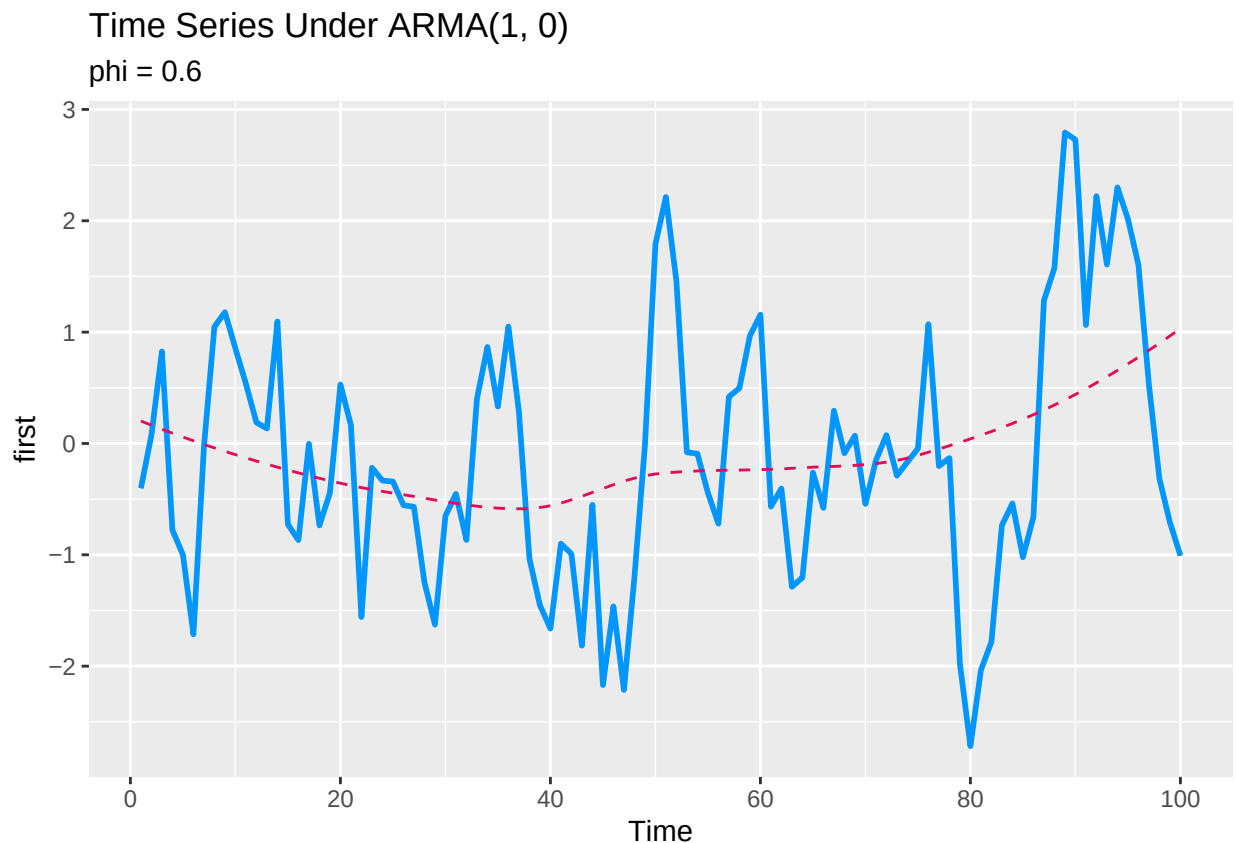
Q2

Recall that the non-seasonal ARIMA is described by three parameters $ARIMA(p, d, q)$ where p is the order of the autoregressive component, d is the number of times the series need to be differenced to obtain stationarity and q is the order of the moving average component. If we don't need to difference the series, we don't need to specify the "I" part and we can use the short version, i.e., the $ARMA(p, q)$.

- (a) Consider three models: $ARMA(1,0)$, $ARMA(0,1)$ and $ARMA(1,1)$ with parameters $\phi = 0.6$ and $\theta = 0.9$. The ϕ refers to the AR coefficient and the θ refers to the MA coefficient. Use the `arma.sim()` function in R to generate $n = 100$ observations from each of these three models. Then, using `autoplot()` plot the generated series in three separate graphs.

```
first <- arma.sim(model = list(order = c(1,
  0, 0), ar = 0.6),
  n = 100)
p1 <- autoplot(first) +
  geom_line(linewidth = 1,
    color = "#0096FF") +
  geom_smooth(linewidth = 0.5,
    color = "#E30B5C",
    linetype = "dashed",
    method = "loess",
    se = FALSE) +
  theme_gray() + labs(title = "Time Series Under ARMA(1, 0)",
    subtitle = "phi = 0.6")
```

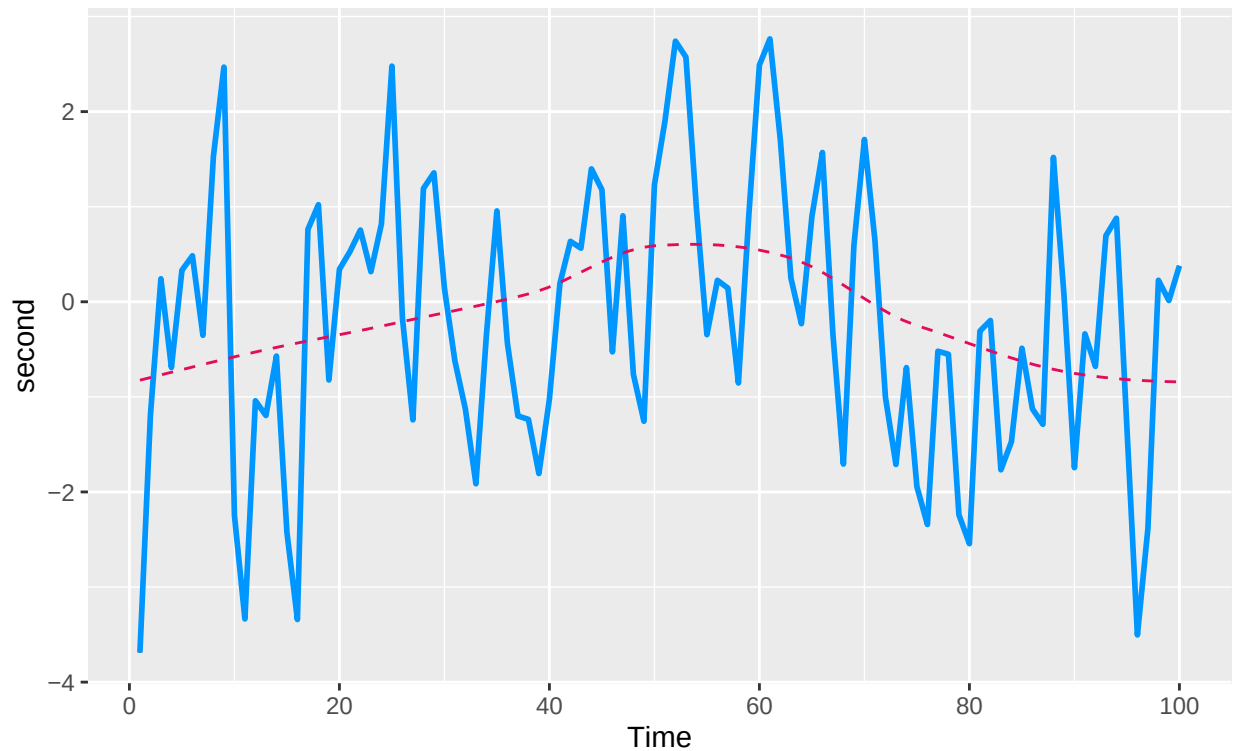
p1



```
second <- arima.sim(model = list(order = c(0,
  0, 1), ma = 0.9),
  n = 100)
p2 <- autoplot(second) +
  geom_line(linewidth = 1,
    color = "#0096FF") +
  geom_smooth(linewidth = 0.5,
    color = "#E30B5C",
    linetype = "dashed",
    method = "loess",
    se = FALSE) +
  theme_gray() + labs(title = "Time Series Under ARMA(0, 1)",
    subtitle = "theta = 0.9")
p2
```

Time Series Under ARMA(0, 1)

theta = 0.9



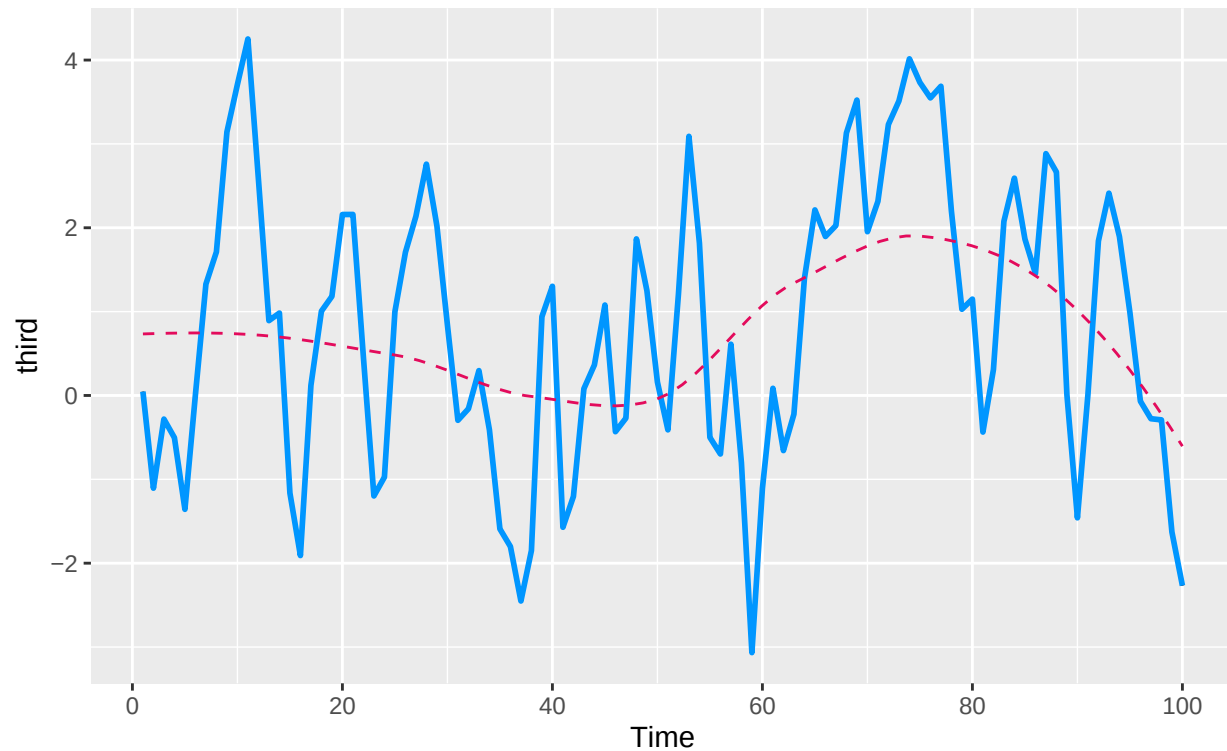
```
third <- arima.sim(model = list(order = c(1,
  0, 1), ar = 0.6,
  ma = 0.9), n = 100)
p3 <- autoplot(third) +
  geom_line(linewidth = 1,
    color = "#0096FF") +
  geom_smooth(linewidth = 0.5,
    color = "#E30B5C",
    linetype = "dashed",
    method = "loess",
```

```
se = FALSE) +
theme_gray() + labs(title = "Time Series Under ARMA(1, 1)",
  subtitle = "phi = 0.6, theta = 0.9")
```

p3

Time Series Under ARMA(1, 1)

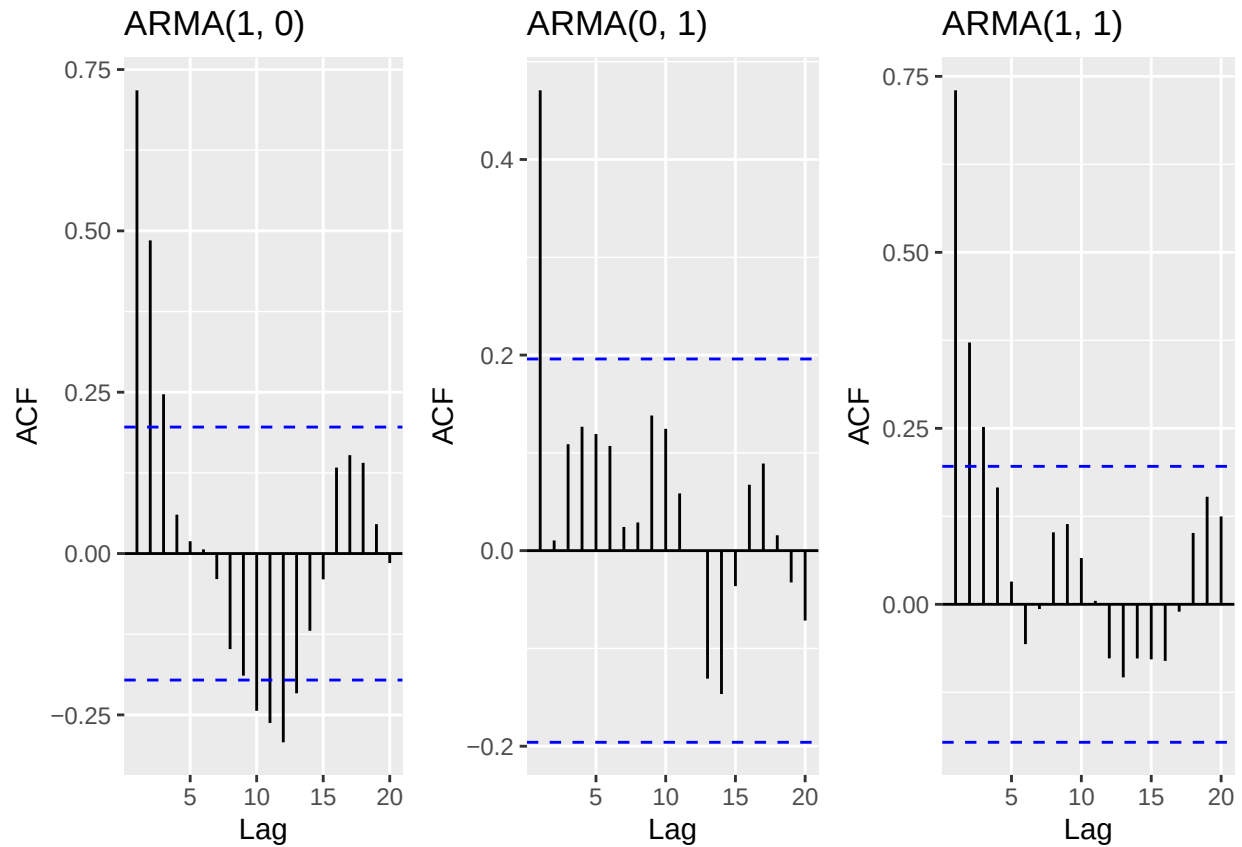
phi = 0.6, theta = 0.9



(b) Plot the sample ACF for each of these models in one window to facilitate comparison (Hint: use `cowplot::plot_grid()`).

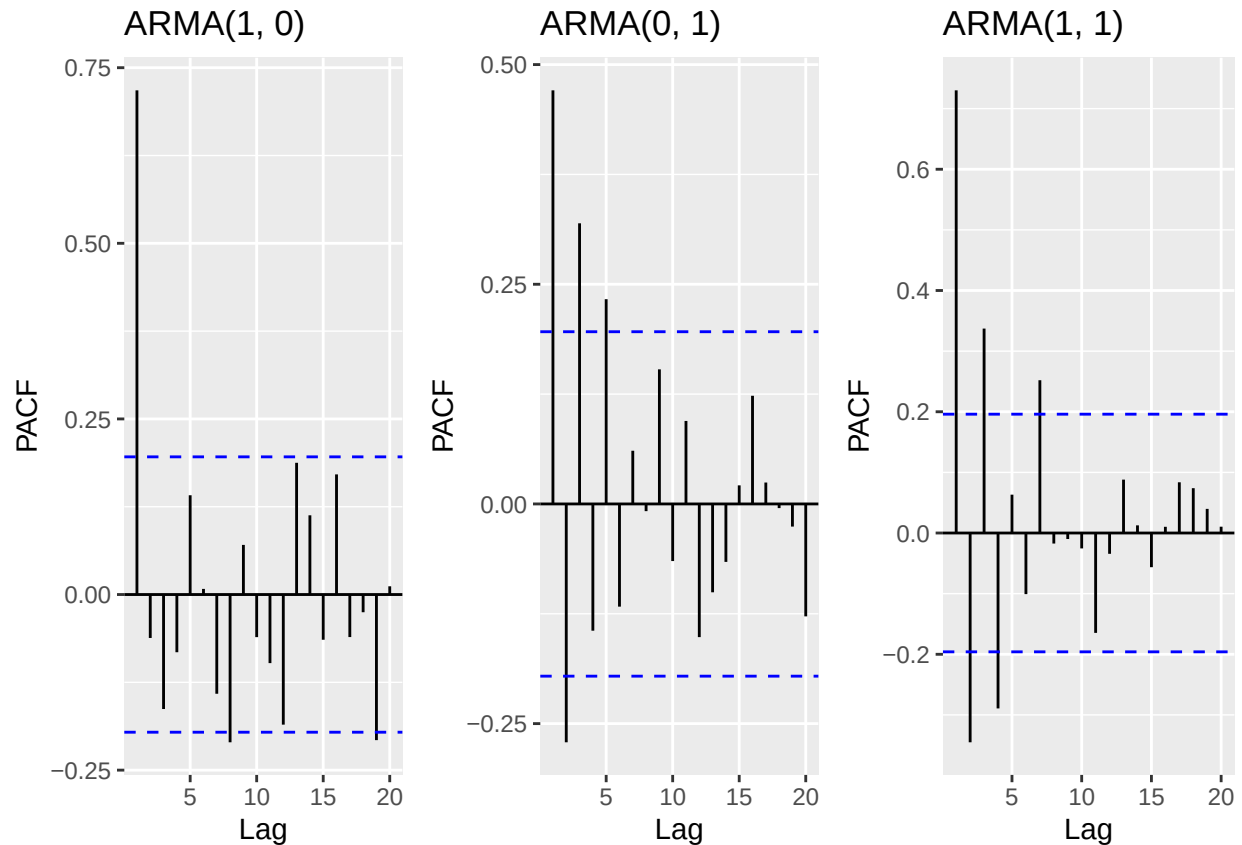
```
p4 <- plot_grid(autoplot(Acf(first,
  lag.max = 20, plot = FALSE),
  main = "ARMA(1, 0)"),
  autoplot(Acf(second,
    lag.max = 20,
    plot = FALSE),
    main = "ARMA(0, 1)"),
  autoplot(Acf(third,
    lag.max = 20,
    plot = FALSE),
    main = "ARMA(1, 1)"),
  nrow = 1)
```

p4



(c) Plot the sample PACF for each of these models in one window to facilitate comparison.

```
p5 <- plot_grid(autoplot(Pacf(first,
  lag.max = 20, plot = FALSE),
  main = "ARMA(1, 0)",
  autoplot(Pacf(second,
    lag.max = 20,
    plot = FALSE),
    main = "ARMA(0, 1)",
    autoplot(Pacf(third,
      lag.max = 20,
      plot = FALSE),
      main = "ARMA(1, 1)",
      nrow = 1)
p5
```



- (d) Look at the ACFs and PACFs. Imagine you had these plots for a data set and you were asked to identify the model, i.e., is it AR, MA or ARMA and the order of each component. Would you be able to identify them correctly? Explain your answer.

Answer: I might identify accurately. For the left AR model, ACF gradually decays, but PACF cuts off with lag 1. For the middle MA model, ACF cuts off with lag 1, but PACF gradually decays. For the right ARMA model, both ACF and PACF gradually decay.

- (e) Compare the PACF values R computed with the values you provided for the lag 1 correlation coefficient, i.e., does $\phi = 0.6$ match what you see on PACF for ARMA(1,0), and ARMA(1,1)? Should they match?

```
Pacf(first, lag.max = 4,
      plot = FALSE)
```

```
##
## Partial autocorrelations of series 'first', by lag
##
##      1      2      3      4
## 0.718 -0.062 -0.163 -0.082
```

```
Pacf(third, lag.max = 4,
      plot = FALSE)
```

```
##
## Partial autocorrelations of series 'third', by lag
##
##      1      2      3      4
## 0.730 -0.345  0.337 -0.289
```

Answer: The results do not match. Theoretically, in a first-order AR model, PACF at lag 1 is equal to 'phi'.

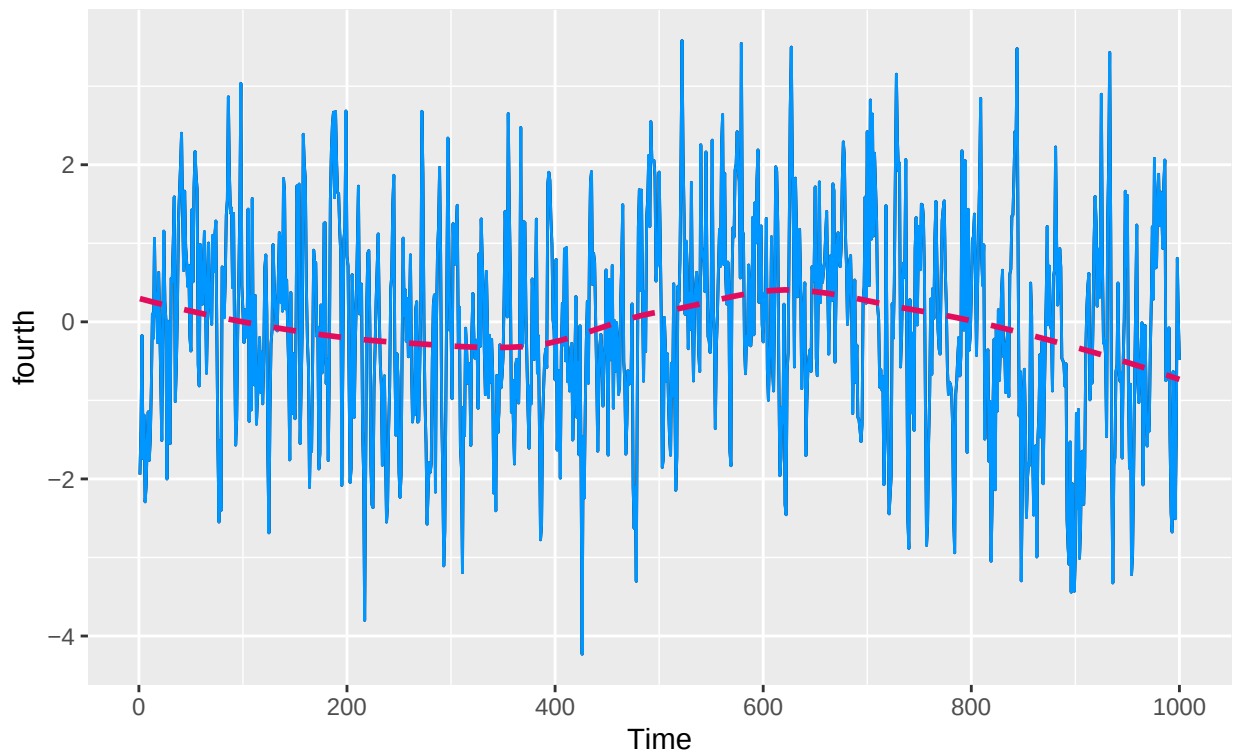
(f) Increase number of observations to $n = 1000$ and repeat parts (b)-(e).

```
fourth <- arima.sim(model = list(order = c(1,
  0, 0), ar = 0.6),
  n = 1000)
p6 <- autoplot(fourth) +
  geom_line(linewidth = 0.5,
    color = "#0096FF") +
  geom_smooth(linewidth = 1,
    color = "#E30B5C",
    linetype = "dashed",
    method = "loess",
    se = FALSE) +
  theme_gray() + labs(title = "Expanded Time Series Under ARMA(1, 0)",
    subtitle = "phi = 0.6")
```

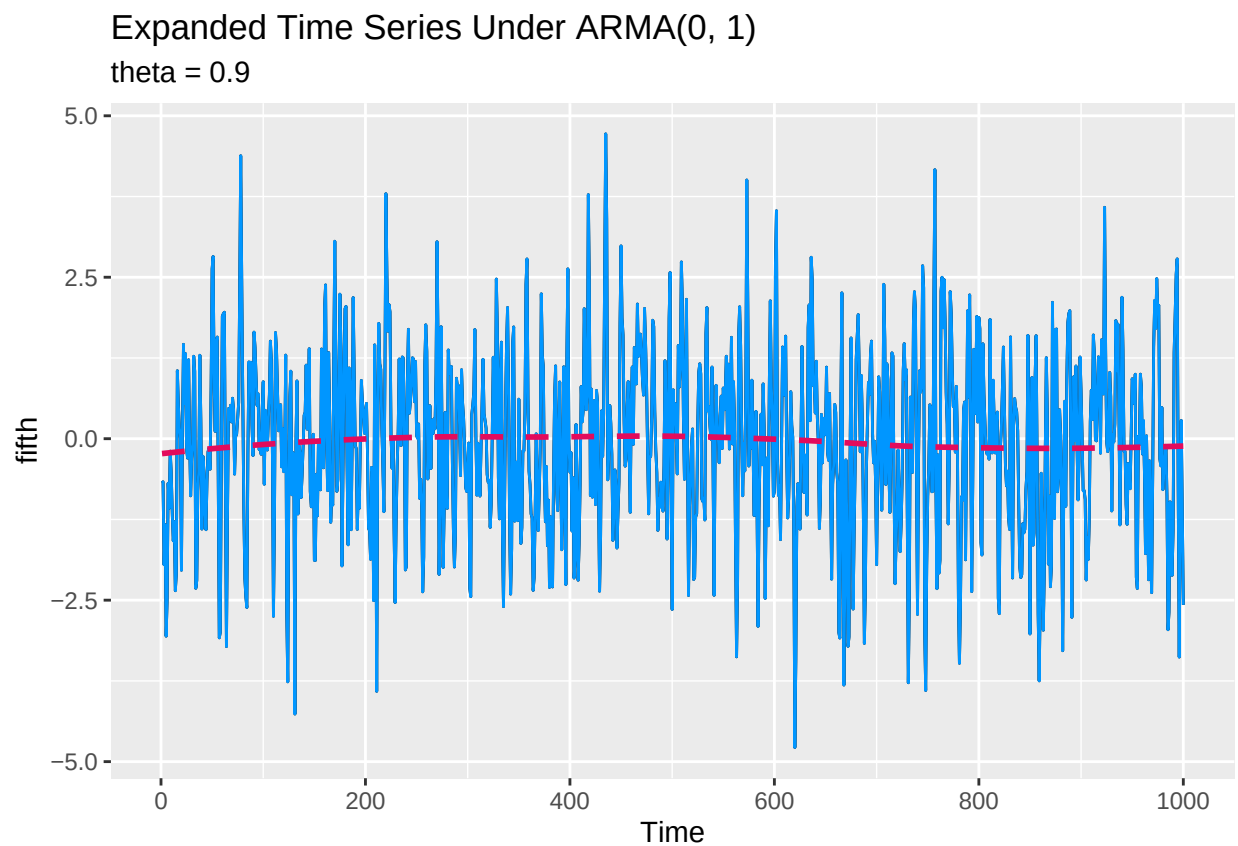
p6

Expanded Time Series Under ARMA(1, 0)

phi = 0.6



```
fifth <- arima.sim(model = list(order = c(0,
  0, 1), ma = 0.9),
  n = 1000)
p7 <- autoplot(fifth) +
  geom_line(linewidth = 0.5,
    color = "#0096FF") +
  geom_smooth(linewidth = 1,
    color = "#E30B5C",
    linetype = "dashed",
    method = "loess",
    se = FALSE) +
  theme_gray() + labs(title = "Expanded Time Series Under ARMA(0, 1)",
    subtitle = "theta = 0.9")
p7
```



```
sixth <- arima.sim(model = list(order = c(1,
  0, 1), ar = 0.6,
  ma = 0.9), n = 1000)
p8 <- autoplot(sixth) +
  geom_line(linewidth = 0.5,
    color = "#0096FF") +
  geom_smooth(linewidth = 1,
    color = "#E30B5C",
    linetype = "dashed",
    method = "loess",
```

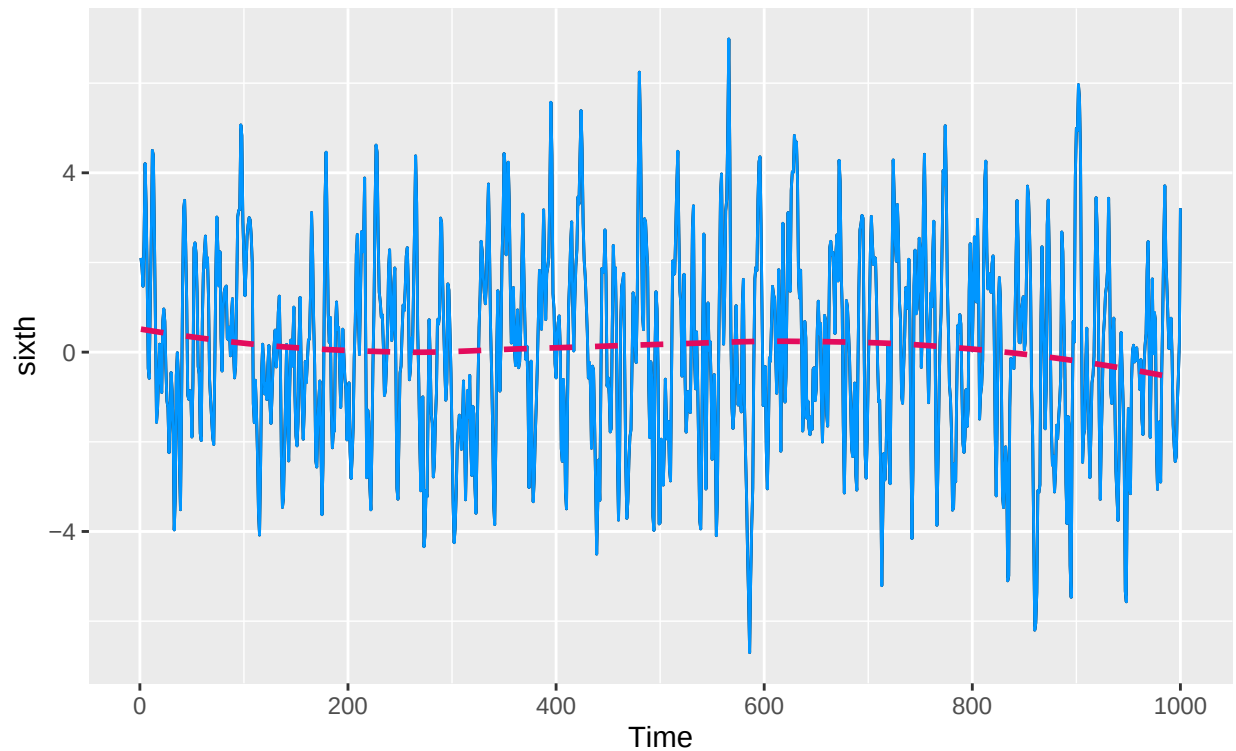


```
se = FALSE) +
theme_gray() + labs(title = "Expanded Time Series Under ARMA(1, 1)",
  subtitle = "phi = 0.6, theta = 0.9")
```

p8

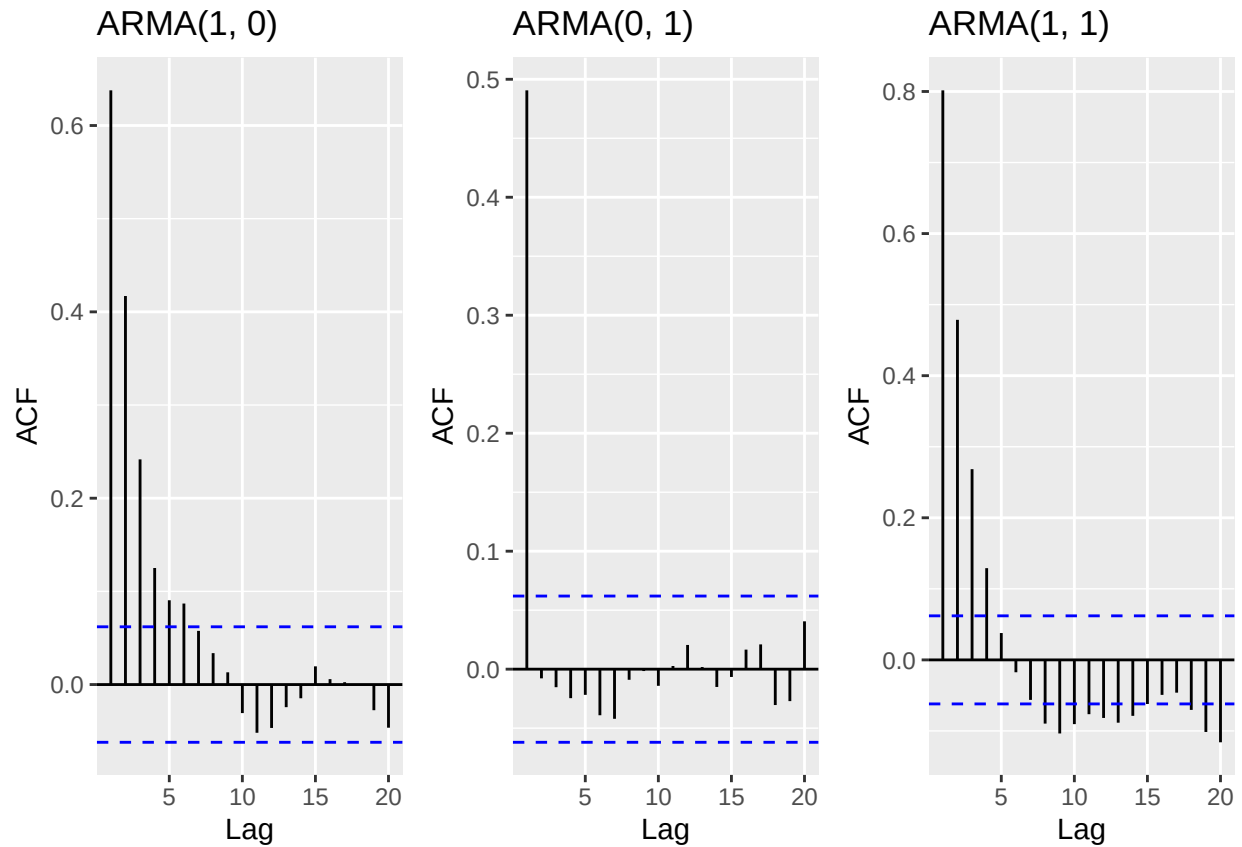
Expanded Time Series Under ARMA(1, 1)

phi = 0.6, theta = 0.9

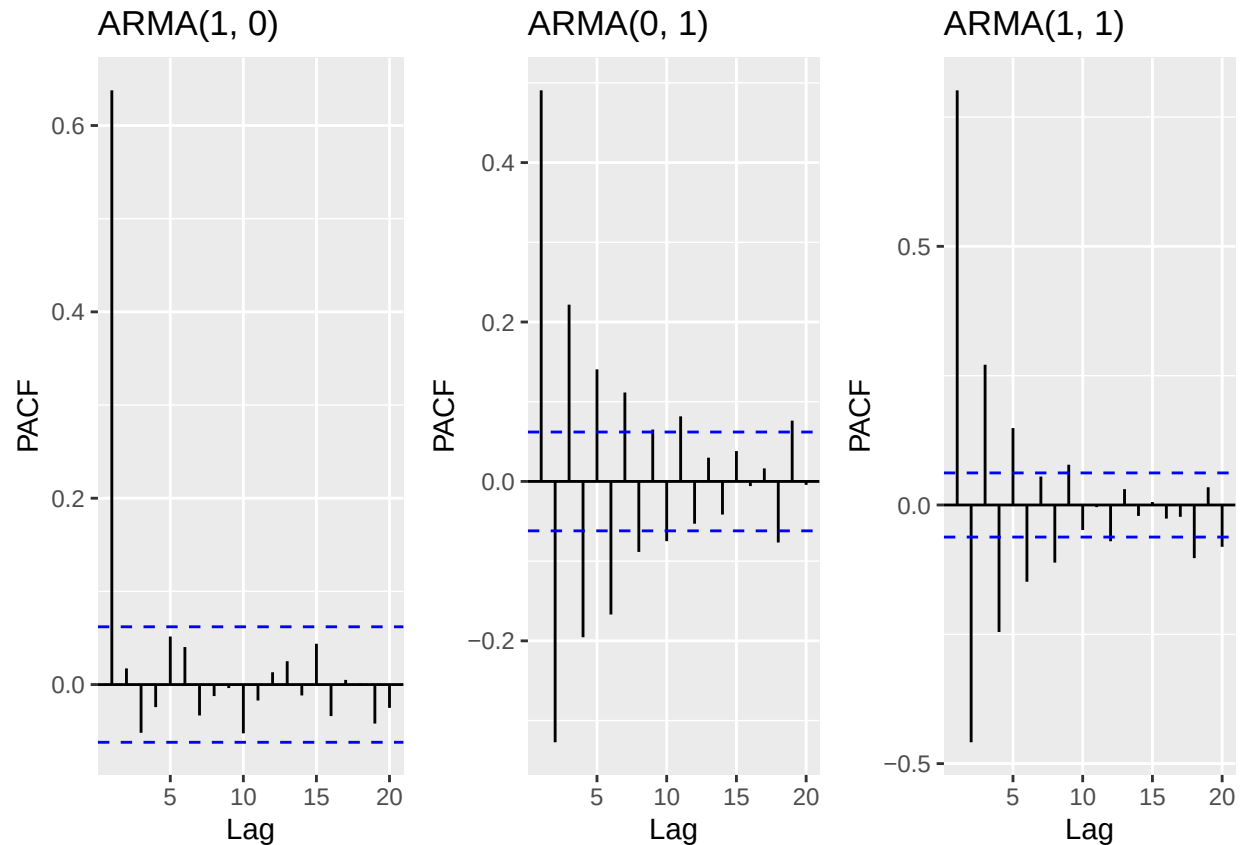


```
p9 <- plot_grid(autoplot(Acf(fourth,
  lag.max = 20, plot = FALSE),
  main = "ARMA(1, 0)"),
  autoplot(Acf(fifth,
    lag.max = 20,
    plot = FALSE),
    main = "ARMA(0, 1)"),
  autoplot(Acf(sixth,
    lag.max = 20,
    plot = FALSE),
    main = "ARMA(1, 1)"),
  nrow = 1)
```

p9



```
p10 <- plot_grid(autoplot(Pacf(fourth,
  lag.max = 20, plot = FALSE),
  main = "ARMA(1, 0)"),
  autoplot(Pacf(fifth,
    lag.max = 20,
    plot = FALSE),
    main = "ARMA(0, 1)"),
  autoplot(Pacf(sixth,
    lag.max = 20,
    plot = FALSE),
    main = "ARMA(1, 1)"),
  nrow = 1)
p10
```



```
Pacf(fourth, lag.max = 4,
     plot = FALSE)
```

```
##
## Partial autocorrelations of series 'fourth', by lag
##
##      1      2      3      4
## 0.638 0.017 -0.052 -0.024
```

```
Pacf(sixth, lag.max = 4,
     plot = FALSE)
```

```
##
## Partial autocorrelations of series 'sixth', by lag
##
##      1      2      3      4
## 0.802 -0.459 0.271 -0.245
```

Answer: I am more confident to recognize three models. Given a longer series, the visual traits might become more obvious. Unfortunately, 'phi' is still different from PACF at lag 1.

Q3

Consider the ARIMA model $y_t = 0.7 * y_{t-1} - 0.25 * y_{t-12} + a_t - 0.1 * a_{t-1}$

- (a) Identify the model using the notation $\text{ARIMA}(p, d, q)(P, D, Q)_s$, i.e., identify the integers p, d, q, P, D, Q, s (if possible) from the equation.

Answer: $p = 1 / d = 0 / q = 1 / P = 1 / D = 0 / Q = 0 / S = 12$

Answer: $\text{SARIMA}(1, 0, 1)(1, 0, 0)[12]$

- (b) Also from the equation what are the values of the parameters, i.e., model coefficients.

Answer: seasonal $\phi = -0.25$ / non-seasonal $\phi = 0.7$ / non-seasonal $\theta = -0.1$

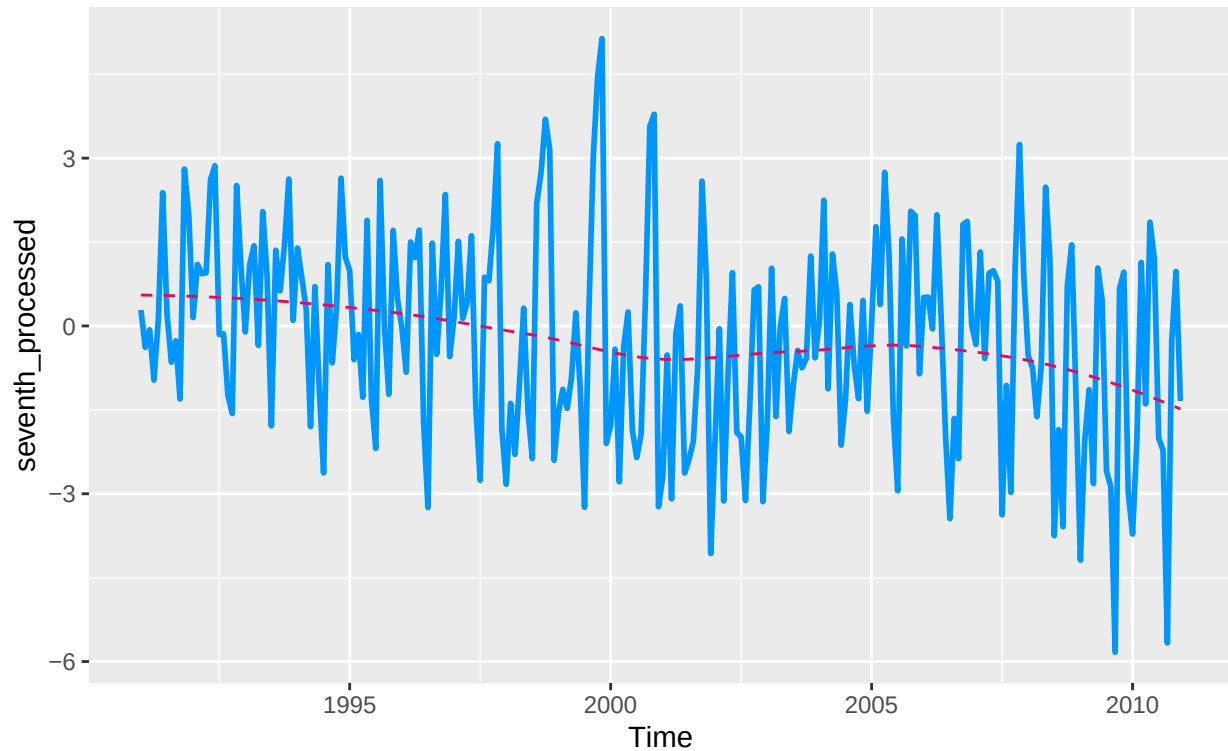
Q4

Simulate a seasonal $\text{ARIMA}(0, 1) \times (1, 0)_{12}$ model with $\phi = 0.8$ and $\theta = 0.5$ using the `sim_sarima()` function from package `sarima`. The 12 after the bracket tells you that $s = 12$, i.e., the seasonal lag is 12, suggesting monthly data whose behavior is repeated every 12 months. You can generate as many observations as you like. Note the Integrated part was omitted. It means the series do not need differencing, therefore $d = D = 0$. Plot the generated series using `autoplot()`. Does it look seasonal?

```
seventh_basic <- sim_sarima(model = list(nseasons = 12,
  sar = 0.8, ma = 0.5),
  n = 240)
seventh_processed <- ts(seventh_basic,
  start = c(1991, 1),
  frequency = 12)
p11 <- autoplot(seventh_processed) +
  geom_line(linewidth = 1,
    color = "#0096FF") +
  geom_smooth(linewidth = 0.5,
    color = "#E30B5C",
    linetype = "dashed",
    method = "loess",
    se = FALSE) +
  theme_gray() + labs(title = "Time Series Under SARIMA(0, 0, 1)(1, 0, 0)[12]",
    subtitle = "seasonal phi = 0.8, non-seasonal theta = 0.5")
p11
```

Time Series Under SARIMA(0, 0, 1)(1, 0, 0)[12]

seasonal phi = 0.8, non-seasonal theta = 0.5

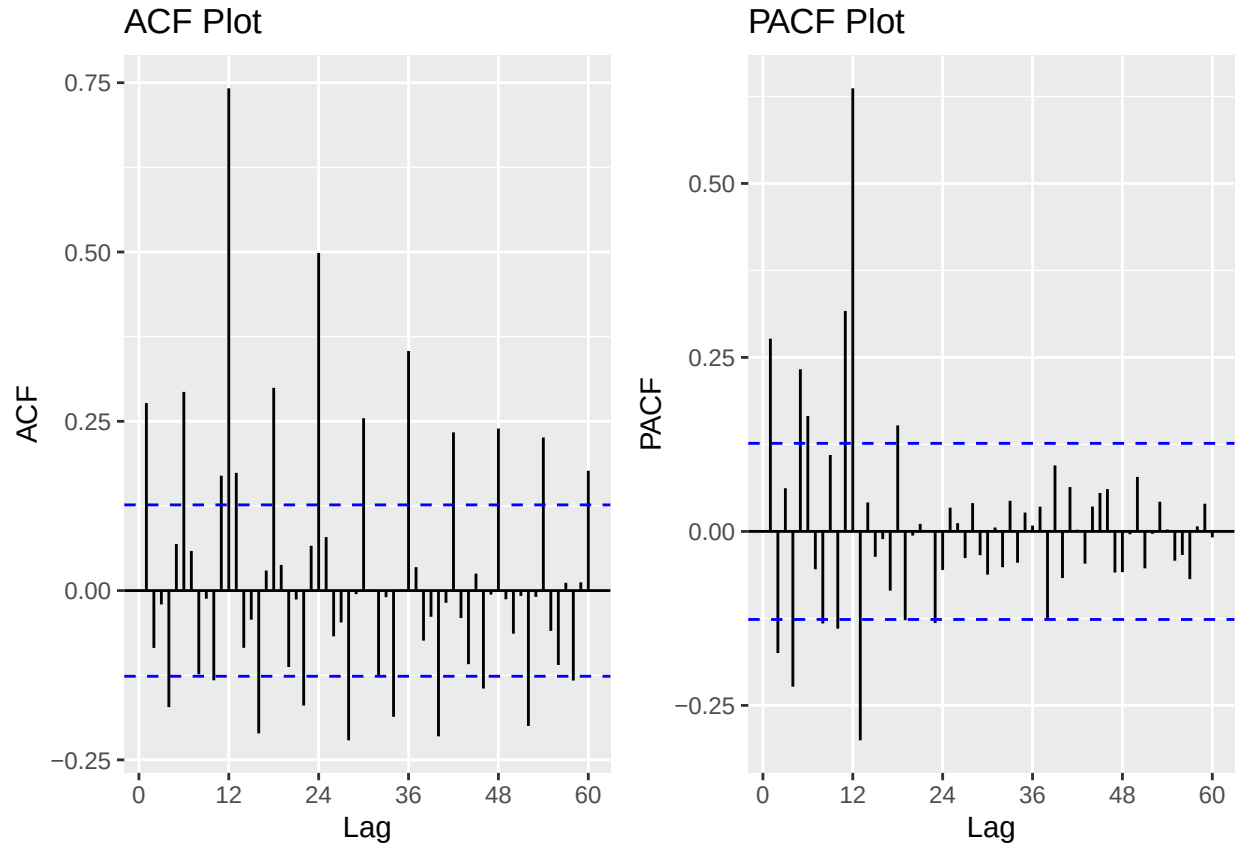


Answer: The interval between two adjacent crests seems to be fixed, so seasonality is confirmed.

Q5

Plot ACF and PACF of the simulated series in Q4. Comment if the plots are well representing the model you simulated, i.e., would you be able to identify the order of both non-seasonal and seasonal components from the plots? Explain.

```
p12 <- plot_grid(autoplot(Acf(seventh_processed,
  lag.max = 60, plot = FALSE),
  main = "ACF Plot"),
  autoplot(Pacf(seventh_processed,
    lag.max = 60,
    plot = FALSE),
    main = "PACF Plot"),
  nrow = 1)
p12
```



Answer: I can discover the property of seasonal AR. In ACF plot, multiple spikes, which carry a downward trend, occur with lag 12, 24, 36, and 48. In PACF plot, lag 12 is associated with the only spike. Therefore, the order is one.

Answer: I can also discover the property of non-seasonal MA. In PACF plot, before lag 12, values gradually decay. In ACF plot, before lag 12, lag 1 is associated with the only spike. Therefore, the order is one.